

CONTACT

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GitHub
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EXPERTISE

Full Stack Development
- React, TypeScript
- Django, Node.js
- MySQL, Redis

DevOps Engineering
- Docker, Kubernetes
- Helm, Harbor, Nginx
- GitHub Actions, CI/CD

AI & Data Engineering
- AI Agent, Tool Use
- N8N, Dify
- Kafka, KubeFlow

PERSONALITY

Detail-oriented:
Focused on robustness and comprehensiveness when architecting system orchestration and scheduling workflows.

Self-motivated:
Passionate about deep-diving into complex technical bottlenecks with long-term research capability.

HONORS

2026 Smart Networks SIG Seminar
AI-RAN commercialization & industry applications · presenter (Sep 2026)

Intl. Workshop on AI-Enabled ICT Networks
6G Research Center, Hanyang University (Korea) · MOE Bilingual Program (2026)

AI-RAN / O-RAN Workshop
National Yang Ming Chiao Tung University (Oct 2025)

Study Abroad Dream Program
SparkAmplify Winter Internship (2026)

Space New Star Program
Industrial Development Admin, MOEA · selected twice (2025, 2026) · matched to OrionGo Technology for indoor-positioning R&D · rated Outstanding in the 2025 practicum

HUANG, YAO-CHING (Kim)

Master's Student / Full Stack & AI Engineer

ABOUT MYSELF

M.S. student in Electronic Engineering at NTUST, focused on **cloud-native platforms**, **MLOps**, **AI agents**, and systems integration. Hands-on experience across platform operations, GPU resource scheduling, automated testing, and fault analysis. Currently contributing to the NSTC 6G AI/ML platform and ITRI's O-RAN digital twin, with a software engineering internship in the US and international technical exchange experience. Seeking roles in **cloud / platform engineering**, **MLOps**, or **AI infrastructure**.

EXPERIENCE

System Operations & Database Administration — NTUST EE Dept. Office
Sep 2026 – Present

- Operate the department's **advisor-matching system** (PHP + MariaDB) and sync enrollment status through the university **student-records API**
- Own **scheduled database backups**, defect fixes, and rebuild / debugging documentation
- Administer the department server room network and day-to-day server operations

Backend / DevOps Engineer — SparkAmplify (California, US)
Jan – Feb 2026

- Designed backend **APIs** and database **schemas**; integrated an **LLM API** for in-product Q&A
- **Containerized deployment**: Docker, **Nginx reverse proxy** with TLS, and a **Harbor** private image registry
- Built a **GitHub Actions CI/CD** pipeline and standardized the team Git workflow and deployment docs

Calculus EMI Teaching Assistant — NTUST
Sep 2025 – Present

- Support **English-medium instruction**, exam design, and student tutoring
- Built an **auto-exam AI agent** that generates complete exam documents via Tool Use with Word
- Developed a grade and exam management system in **React + Django**

Server Administration (MIS / Network Lead) — Ying-Na Industrial Co.
Aug 2025 – Present

- Maintain company servers, databases, and internal systems, including **backup management**
- Handle IT equipment procurement and **software licence** renewals
- Troubleshoot network connectivity and hardware / software issues

KEY PROJECTS

6G AI/ML Training Platform — NSTC Four-Year Research Program
Jun 2025 – Present

A shared AI/ML training platform; owned platform engineering in year 3 and AI agent architecture in year 4.

Year 3 **Platform Eng.**	Operate **20+ container services** and **Kubeflow / Argo** pipelines with **Prometheus + DCGM** monitoring and **9 repeatable** load-test suites; **root-caused and fixed 7 platform faults**, cutting image-build failure under load **27% → 0%** and raising the single-object ceiling **470 MB → 4.7 GB**.
Year 3 **GPU Scheduling**	Implemented a **non-invasive GPU allocator** outside the pipeline engine (resource gateway, ledger, size-aware exclusive-or-shared placement) with **NVIDIA MPS** partitioning: throughput **+21–24%**, batch makespan **−13%**.
Year 4 **AI Agent**	Designed a **multi-agent** architecture: a training agent and a monitoring agent in an iterative loop, with intent, workflow, and memory defined in the **system prompt**; collapsed **17 low-level APIs into 10 Skills** behind an interface-translation layer, **sharply reducing multi-step operation errors**.

Kubernetes · Docker · Kubeflow · Argo · NVIDIA MPS · DCGM · Prometheus · Dify

5G NR RAN Digital Twin — ITRI Industry-Academia Collaboration
Apr 2026 – Present

Rebuilt a base-station environment on commodity hardware so control algorithms can be validated before field deployment.

Simulation	Built an **O-RAN RAN simulator**, combining **NVIDIA Omniverse** and **Sionna RT** for 3D scene and wireless channel simulation.
Integration	Developed the interface layer between the simulator and a **Near-RT RIC** so **xApps switch between simulated and physical environments without code changes**.
Validation	Built the **KPI monitoring frontend**; aligned primary metrics to **90–96%** against **OAI rfsim** and cut mean error **50.1% → 2.5%** with an **ML calibrator**.

O-RAN · 5G NR · OpenAirInterface · Sionna RT · NVIDIA Omniverse · Django

EDUCATION

National Taiwan University of Science and Technology (NTUST)
Sep 2025 – Jun 2027

M.S. in Electronic Engineering

Mobilization Information Technology Lab | Advisor: Prof. Jenq-Shiou Leu | GPA: 4.18 / 4.30

National Taiwan University of Science and Technology (NTUST)
Sep 2021 – Jun 2025

B.S. in Electronic Engineering

- GPA: 4.04 / 4.30 | Class Rank: 3 / 29 (Top 10.34%) | Dept. Rank: 10 / 104 (Top 9.62%)

RESEARCH & THESIS

SSD Tail Latency Optimization — DQN-Based Idle-Time GC Scheduling
B.S. Thesis

Applied Deep Q-Learning for RL-based GC scheduling; validated with 100K SNIA IOTTA traces on MQSim. Bridged Python/C++ via Cython; effectively reduced long tail latency occurrence.